

Subham Bhattacharya

Backend Engineer · Python

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Backend engineer focused on Python, with merged contributions to two widely used Python open-source projects and project experience in multi-tenant data isolation, role-based access control and asynchronous I/O under strict type checking. Computer Science undergraduate at BITS Pilani.

TECHNICAL SKILLS

Languages Python, TypeScript, SQL, C, HTML/CSS

Backend Flask, REST API design, PostgreSQL, SQLite, relational data modelling, multi-tenant architecture, session authentication, RBAC, CSRF protection, password hashing, HTTP semantics, Jinja2

Async & Concurrency `asyncio`, `httpx`, bounded concurrency, per-host rate limiting, retries and backoff, timeouts, `robots.txt` compliance

Typing, Testing & Quality `mypy --strict`, Protocols, generics, dataclasses, generators, decorators, context managers, iterators, `pytest`, regression testing, `ruff`, ESLint, pre-commit hooks, GitHub Actions, CI, code review

Infrastructure & Web Git, Linux, Bash, uv, Cloudflare Pages/Workers, GraphQL, Content-Security-Policy, Next.js, React, Tailwind CSS, Zod

OPEN SOURCE CONTRIBUTIONS

Litestar

[PR #4990](#) — merged

High-performance Python ASGI web framework

- Corrected HTTP semantics for multipart form part-limit violations, returning `413 Request Entity Too Large` in place of `400`; shipped with regression tests, a changelog entry and a breaking-change note. Reviewed and merged by a core maintainer.

Strawberry GraphQL

[PR #4591](#) · [#4581](#) — merged

GraphQL library for Python

- Corrected Strawberry's `dataclass_transform` metadata (PEP 681) to declare `order_default=False`, aligning static type-checker behaviour with runtime dataclass semantics — the previous value made `mypy` and `ty` reject valid user-defined comparison methods. Applied across related decorators with regression coverage; separately updated resolver docs to be independently runnable.

PROJECTS

Postmark — AI-Powered Email Threat Detection Platform

[Demo](#)

Python · FastAPI · `async services` · PostgreSQL (Smart India Hackathon 2026 — selected)

- Team lead** of six engineers selected for Smart India Hackathon 2026 (SIH26106) — email threat detection with geolocation and forensic intelligence. Owned backend architecture and split delivery across backend, DevOps, ML and frontend.
- Built the backend in FastAPI: `async` request handling, database schema and access layer, and the test and git workflows the team shipped against. Repository private by design.

site-auditor — Concurrent Crawler & Link-Health Auditor

[Code](#)

Python 3.12 · `asyncio` · `httpx` · `mypy --strict` (in development)

- Async network layer complete: a pooled `httpx.AsyncClient` under explicit limits, retry with exponential backoff and jitter (policy as data), per-host request spacing, and `robots.txt` parsed and honoured — behind Protocol-defined interfaces.
- Test doubles satisfy the `Fetcher` protocol structurally, not by inheritance, so the retry path runs against a real failing-then-succeeding fetcher with no mocking library. Built phase by phase through reviewed PRs under `mypy --strict` and `ruff`, green CI throughout.

OrderFlow — Multi-Tenant B2B Order Management

[Code](#) · [Demo](#)

Python · Flask · SQLite · Jinja2 · GitHub Actions

- Shared schema with every query scoped by `company_id`, so no tenant reads or writes across an organisational boundary; role separation via `admin_required` / `client_required` decorators, server-side sessions, password hashing and CSRF protection.
- Modelled the order lifecycle as an explicit state machine (`pending` → `approved` → `fulfilled` / `rejected`), validating each transition server-side rather than trusting client state.
- Represented currency as integer paise rather than floating point, eliminating rounding drift across totals and invoices.

PROFESSIONAL EXPERIENCE

Founder & Managing Director

May 2024 — Present

Agam Agrovet Private Limited

Kolkata, West Bengal

- End-to-end ownership of an animal nutrition manufacturing business: production scheduling, quality control, procurement across 40+ suppliers, distribution to 60+ customers.
- Replaced a manual register process with digital purchase-order tracking, cutting procurement turnaround 40%; automated payables and receivables tracking, cutting overdue payments 40%.

Managing Director

Sep 2022 — Present

Anandam Breeding Private Limited

Kolkata, West Bengal

- Lead day-to-day operations, supply chain, inventory control and distribution for an animal health products business, managing a 20-person team.

EDUCATION & CERTIFICATIONS

BSc (Hons), Computer Science, BITS Pilani · Semester 1 GPA 10.00/10.00 ([grade sheet](#)) · Mar 2026 — Mar 2030

Certifications CS50x, Harvard · Python for Everybody, U. Michigan · Open Source Community (LFC102), Linux Foundation · CS50 AI (in progress)

Coursework Computing Systems (C, pointers, memory) · Discrete Mathematics · Linear Algebra | **Languages** English · Bengali · Hindi